Confidence intervals, pivotal quantities

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Recap

Confidence set: Let $\theta \in \Theta$ be a parameter of interest, and $X_1, ..., X_n$ a sample. A set $C(X_1, ..., X_n) \subseteq \Theta$ is a $1 - \alpha$ **confidence set** for θ if

$$\inf_{\theta \in \Theta} P_\theta(\theta \in C(X_1, ..., X_n)) = 1 - \alpha$$

Inverting a test: Create a confidence set by inverting a test:

$$C(X_1, ..., X_n) = \{\theta_0 : (X_1, ..., X_n) \notin \mathcal{R}(\theta_0)\}$$

Using confidence sets to test hypotheses

<u>Theorem</u>: Let $\theta \in \Theta$ and let $C(X_1, \dots, X_n)$ be a $1-\alpha$ confidence set.

For any $\theta_0 \in \Theta$, let

$$R(\theta_0) = \{ (X_1, \dots, X_n) : \theta_0 \notin C(X_1, \dots, X_n) \}$$

The test that rejects $H_0: \theta = \theta_0$ when $(X_1, \dots, X_n) \in R(\theta_0)$ is a level α test

<u>Pf</u>: WTS $\forall \theta_0 \in \Theta$, $P_{\theta_0}((X_1, \dots, X_n) \in R(\theta_0)) \leq \alpha$

So, let $\theta_0 \in \Theta$. Then

$$P_{\theta_0}((X_1, \dots, X_n) \in R(\theta_0)) = P_{\theta_0}(\theta_0 \notin C(X_1, \dots, X_n))$$

$$= 1 - \underbrace{P_{\theta_0}(\theta_0 \in C(X_1, \dots, X_n))}_{\geq 1-\alpha}$$

$$\leq 1 - (1-\alpha) = \alpha$$

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t interval

Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$. We want to construct a $1-\alpha$ CI for μ.

$$\underbrace{Q(X_1, \dots, X_n, \mu)}_{\text{pivotal quantity}} = \frac{\sqrt{n}(\bar{X} - \mu)}{S} \sim t_{n-1}$$

does not depend on μ

To construct a confidence set for μ:

Find a, b st

$$P_\mu(a \le Q(X_1, \dots, X_n, \mu) \le b) = 1 - \alpha$$

In this example, $Q(X_1, \dots, X_n, \mu) \sim t_{n-1}$

$b = t_{n-1, \frac{\alpha}{2}}$

$a = -t_{n-1, \frac{\alpha}{2}}$

t interval

Suppose $X_1, ..., X_n \overset{iid}{\sim} N(\mu, \sigma^2)$. We want to construct a $1-\alpha$ CI for μ.

$$Q(X_1, \dots, X_n, \mu) = \frac{\sqrt{n}(\bar{X}-\mu)}{S} \sim t_{n-1}$$

$$P_\mu\left(-t_{n-1,\frac{\alpha}{2}} \le Q(X_1, \dots, X_n, \mu) \le t_{n-1,\frac{\alpha}{2}}\right) = 1-\alpha$$

don't depend on μ

confidence set: $\{\mu : Q(X_1, \dots, X_n, \mu) \in [-t_{n-1,\frac{\alpha}{2}}, t_{n-1,\frac{\alpha}{2}}]\}$

$$-t_{n-1,\frac{\alpha}{2}} \le \frac{\sqrt{n}(\bar{X}-\mu)}{S} \le t_{n-1,\frac{\alpha}{2}}$$

$$\iff \bar{X} - t_{n-1,\frac{\alpha}{2}} \frac{S}{\sqrt{n}} \le \mu \le \bar{X} + t_{n-1,\frac{\alpha}{2}} \frac{S}{\sqrt{n}}$$

So; confidence set $= \bar{X} \pm t_{n-1,\frac{\alpha}{2}} \frac{S}{\sqrt{n}}$

Pivotal quantities

Q is <u>not</u> a statistic (b/c θ is unknown)

Let $X_1, ..., X_n$ be a sample and θ be an unknown parameter. A function $Q(X_1, ..., X_n, \theta)$ is called a **pivot** if the distribution of $Q(X_1, ..., X_n, \theta)$ does not depend on θ.

To get a $1-\alpha$ confidence set for θ

① Find a, b st $P_\theta(a \le Q(X_1, ..., X_n, \theta) \le b) = 1-\alpha$

(can do this b/c distribution of Q does not depend on θ)

② $1-\alpha$ confidence set:

$$\{\theta : a \le Q(X_1, ..., X_n, \theta) \le b\}$$

(often, try to rearrange to get a nice interval for θ, like in t example)

Example

Let $X_1, ..., X_n \overset{iid}{\sim} Uniform(0, \theta)$. We want to construct a $1 - \alpha$ CI for θ using a pivotal quantity.

$\hat{\theta} = X_{(n)}$, so maybe we can use $X_{(n)}$ to create a confidence set

$$P_\theta(X_{(n)} \le t) = P_\theta(X_1 \le t, \dots, X_n \le t)$$

$$= (P_\theta(X_i \le t))^n = \left(\frac{t}{\theta}\right)^n \qquad t \in [0, \theta]$$

$$Q(X_1, \dots, X_n, \theta) = Q(X_{(n)}, \theta) = \frac{X_{(n)}}{\theta}$$

$$P_\theta(Q(X_{(n)}, \theta) \le t) = P\left(\frac{X_{(n)}}{\theta} \le t\right) = P(X_{(n)} \le \theta t)$$

$$= \left(\frac{\theta t}{\theta}\right)^n = t^n$$

$t \in [0, 1]$

no θ!

$\Rightarrow \quad Q(X_{(n)}, \theta) = \dfrac{X_{(n)}}{\theta}$

is a pivot!

Example

Let $X_1, ..., X_n \overset{iid}{\sim} Uniform(0, \theta)$. We want to construct a $1 - \alpha$ CI for θ using a pivotal quantity.

$$Q(X_{(n)}, \theta) = \frac{X_{(n)}}{\theta}$$

Find a, b st $P_\theta\left(a \le \frac{X_{(n)}}{\theta} \le b\right) = 1 - \alpha$

$$P_\theta\left(\frac{X_{(n)}}{\theta} \le t\right) = t^n$$

$$\Rightarrow P_\theta\left(a \le \frac{X_{(n)}}{\theta} \le b\right) = b^n - a^n \qquad a, b \in [0,1]$$

So we want a, b st $b^n - a^n = 1 - \alpha$

E.g. $b = 1 \Rightarrow 1 - a^n = 1 - \alpha \Rightarrow a = \alpha^{1/n}$

$$\Rightarrow P_\theta\left(\alpha^{1/n} \le \frac{X_{(n)}}{\theta} \le 1\right) = 1 - \alpha$$

$$\Rightarrow P_\theta\left(X_{(n)} \le \theta \le \frac{X_{(n)}}{\alpha^{1/n}}\right) = 1 - \alpha$$

$$CI: \left[X_{(n)}, \frac{X_{(n)}}{\alpha^{1/n}}\right]$$

Example

Let $X_1, ..., X_n \overset{iid}{\sim} Exponential(\theta)$, with pdf $f(x|\theta) = \theta e^{-\theta x}$. Find a pivotal quantity $Q(X_1, ..., X_n, \theta)$ and construct a $1 - \alpha$ confidence interval for θ using that quantity.

Hints:

- If $X \sim Exponential(\theta)$, then $cX \sim Exponential\left(\frac{\theta}{c}\right)$
- The sum of exponential random variables has a Gamma distribution

MLE: $\hat{\theta} = \frac{n}{\sum_{i=1}^{n} X_i}$ Suggests we might consider $\sum_i X_i$

$X_i \overset{iid}{\sim} \text{Exponential}(\theta) \Rightarrow \theta X_i \overset{iid}{\sim} \text{Exponential}(1)$

$Q(X_1, \dots, X_n, \theta) = \theta \sum_{i=1}^{n} X_i \sim \text{Gamma}(n, 1)$

pivot

distribution does not depend on θ!

$$Q(X_1, \dots, X_n, \theta) = \theta \sum_{i=1}^{n} X_i \sim \text{Gamma}(n, 1)$$

Next, find a, b st $P_\theta\left(a \le \theta \sum_{i=1}^{n} X_i \le b\right) = 1-\alpha$

E.g.: let $F_{n,1}$ be Gamma$(n,1)$ cdf

$$a = F_{n,1}^{-1}\left(\frac{\alpha}{2}\right) \qquad b = F_{n,1}^{-1}\left(1 - \frac{\alpha}{2}\right)$$

$$P_\theta\left(a \le \theta \sum_{i=1}^{n} X_i \le b\right) = 1-\alpha$$

$$\Rightarrow P_\theta\left(\frac{a}{\sum_{i=1}^{n} X_i} \le \theta \le \frac{b}{\sum_{i=1}^{n} X_i}\right) = 1-\alpha$$

$1-\alpha$ CI: $\left[\dfrac{a}{\sum_{i=1}^{n} X_i}, \dfrac{b}{\sum_{i=1}^{n} X_i}\right]$